OCEAN HACKATHON 2026
CHENNAI EDITION

PROBLEM STATEMENT TITLE - Sea Fusion Energy & Water Hub

THEME - Integrated Ocean Renewable Energy and Sea Water Desalination Solution

CHE CATEGORY - CHE02(Hardware)

LOCATION - Chennai

TEAM NAME - WHALE



PROBLEM

- Freshwater scarcity in coastal and remote island communities.
- High electricity costs and dependence on diesel generation.
- Energy-intensive desalination and maintenance challenges.
- Underutilisation of integrated ocean-based renewable resources.

OUR IDEA

- Combine OTEC power generation and LTTD in one integrated hub.
- Optimise seawater intake and heat-exchange infrastructure.
- Use generated electricity to support desalination and auxiliary equipment.
- Design a modular system adaptable to islands and coastal communities.

PROPOSED SOLUTION

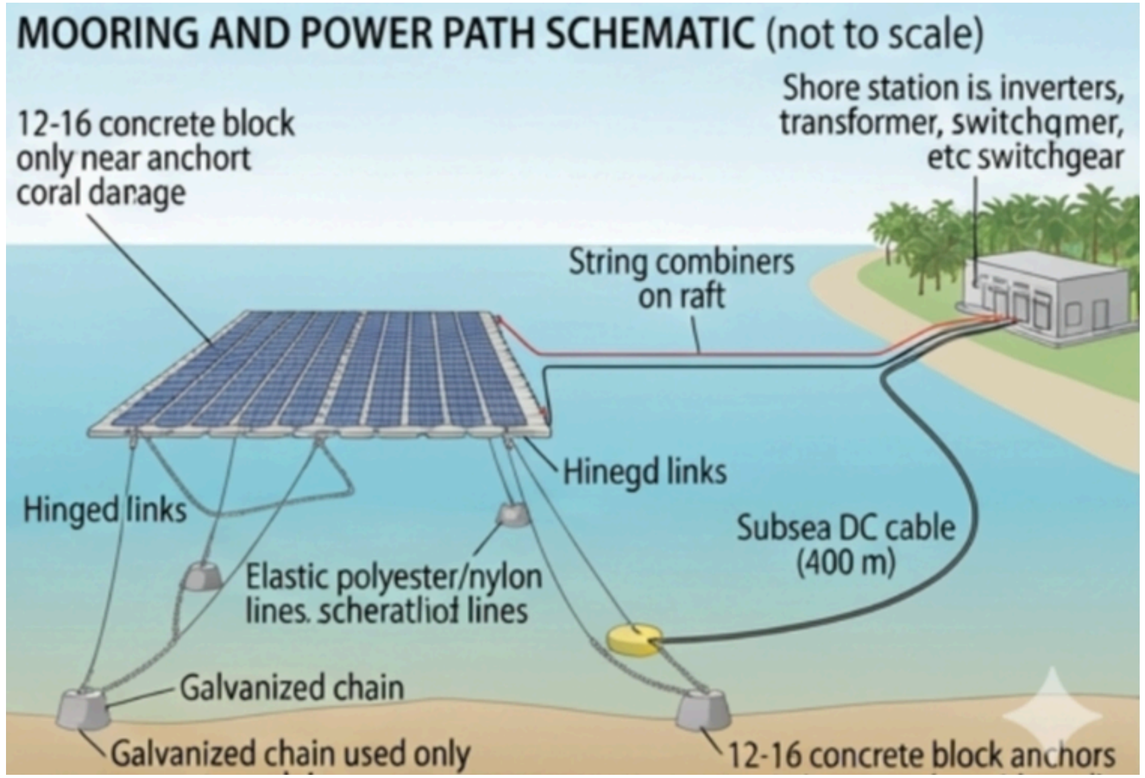
- Integrate OTEC with LTTD desalination.
- Generate electricity using the ocean's thermal gradient.
- Produce freshwater through low-temperature thermal desalination.
- Develop a scalable, solar, wind and wave and water system.

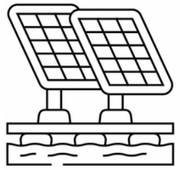
UNIQUENESS

- Dual output of renewable electricity and freshwater.
- Efficient integration of ocean thermal energy and desalination.
- Potential reduction in external energy demand and operating costs.
- Scalable design for diverse island applications.

SOLAR ENERGY

- **Technology:** Deploy floating photovoltaic (FPV) panels on a sheltered coastal waterbody.
- **Energy generation:** Convert sunlight into electrical energy using solar PV modules.
- **System design:** Use modular floating rafts, marine-grade mounting structures and corrosion-protected electrical connections.
- **Integration:** Supply renewable electricity to the proposed energy hub, including desalination auxiliaries and other loads.
- **Installation cost:** Rs.2.5 crore





214 kWp
Proposed capacity
Installed solar power



879 kWh/day
Estimated generation
Daily electrical energy



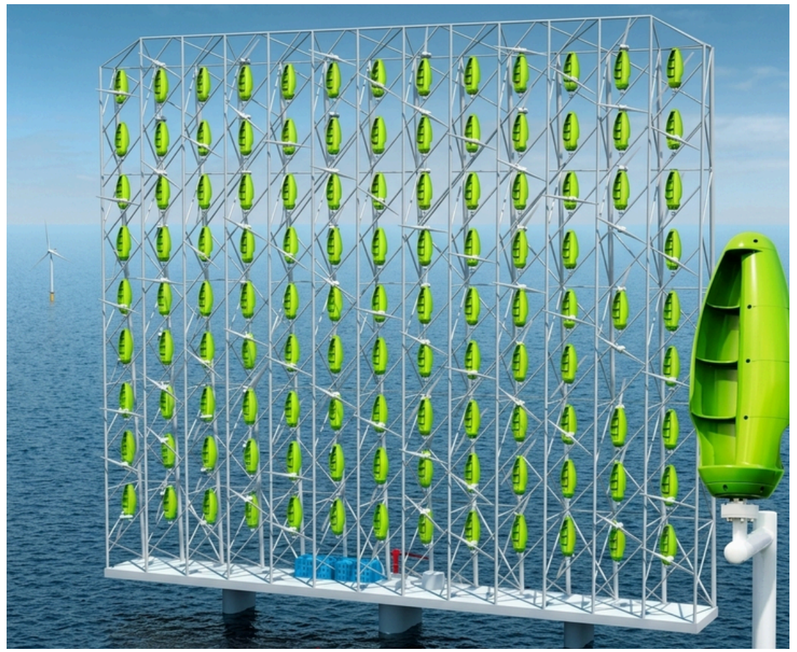
2,200–2,450 m²
Array footprint
Including spacing



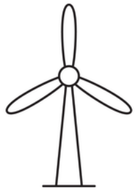
Nearshore
Deployment zone
Sheltered waters

WIND ENERGY

- **Renewable Generation**: Harness coastal and offshore winds to generate electricity.
- **Proposed Technology**: Explore compact vertical-axis wind turbines, such as Aeroleaf units, for modular deployment.
- **System Integration**: Combine wind-generated electricity with solar and OTEC power to support desalination and auxiliary loads.
- **Marine Feasibility**: Assess wind conditions, corrosion resistance, structural stability, maintenance, and storm survival before deployment.
- **Installation cost:** Rs.1.1crore



***AI generated(For Visualization)**



100 kW
Proposed capacity
Rated wind power



360 kWh/day
Average daily generation
Estimated electrical energy



15%
Implied capacity factor



Nearshore
Deployment zone
Sheltered waters

WAVE ENERGY

- **Problem**: Coastal waves contain renewable energy that remains underutilised in many island communities.
- **Technology**: Use a two-body point-absorber wave energy converter that captures relative motion between floating bodies.
- **Energy conversion**: Convert wave-induced motion into electricity through a power take-off (PTO) mechanism and generator.
- **Integration**: Feed the generated electricity into the hybrid energy system to support desalination and auxiliary loads.



45 kW

Rated power

As specified: per wave crest

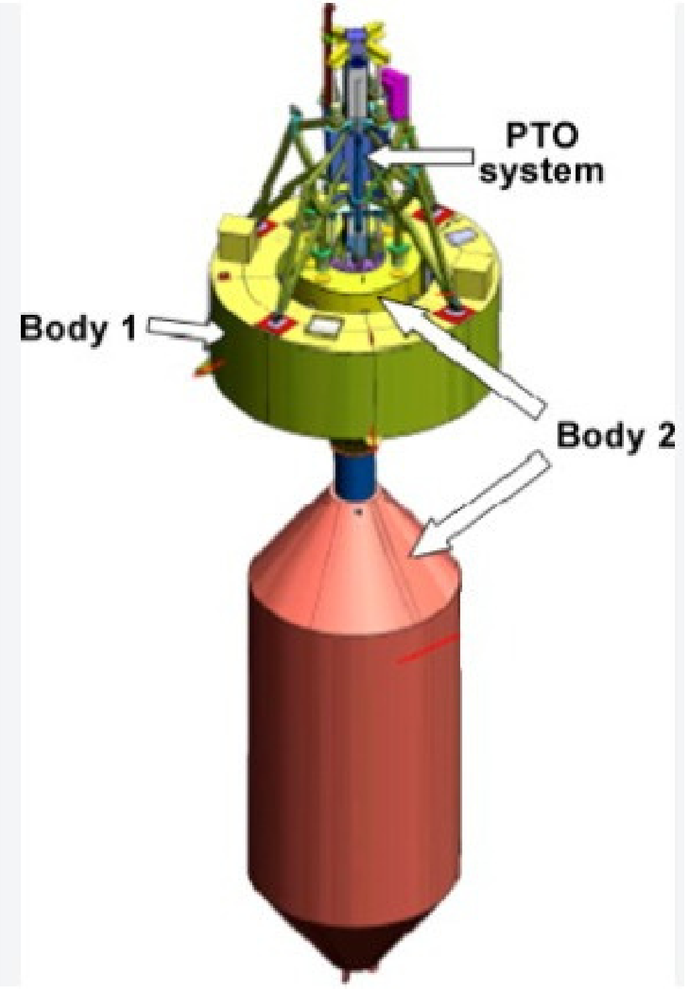


2500 kWh/day

Daily energy generation

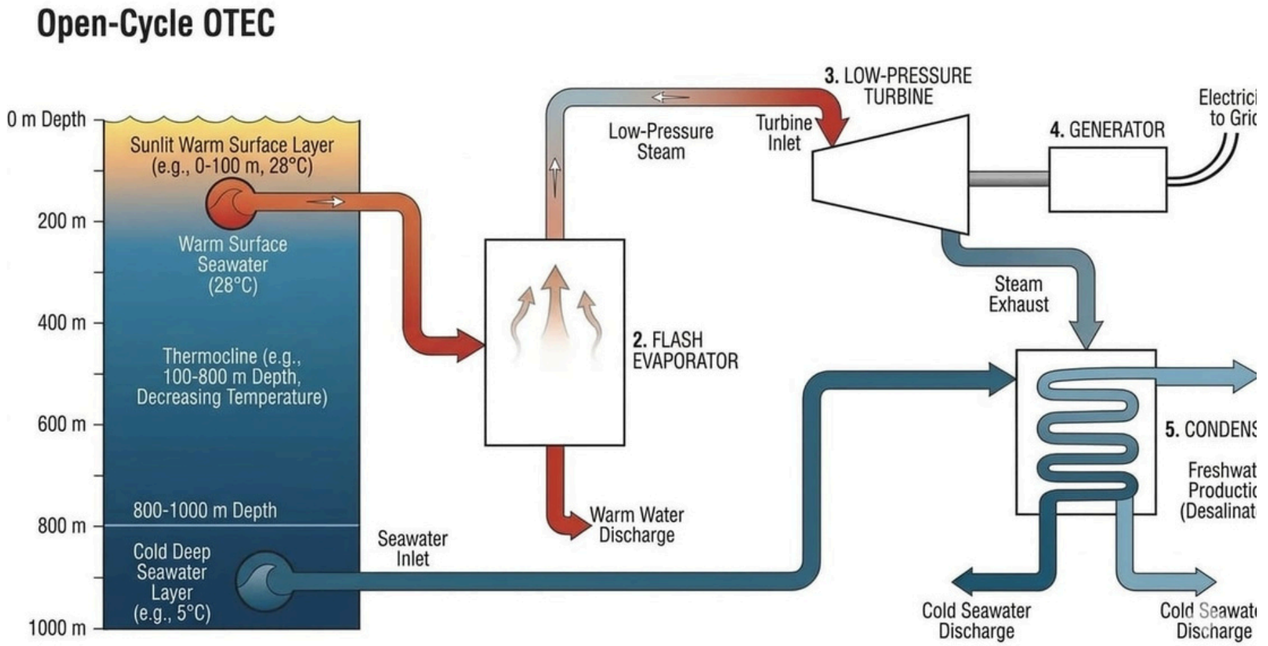
Estimated electrical energy

Location	Wave power	Capacity factor	Average power	Energy per day
Inside lagoon	0.05-0.15 kW/m	about 3-8%	0.5-1.6 kW	about 12-37 kWh
East coast	2-6 kW/m	about 15-25%	3-5 kW	about 72-120 kWh
West/south open sea	8-12 kW/m (28+ in monsoon)	about 25-35%	5-7 kW	about 120-170 kWh



OPEN CYCLE Ocean Thermal Energy Conversion

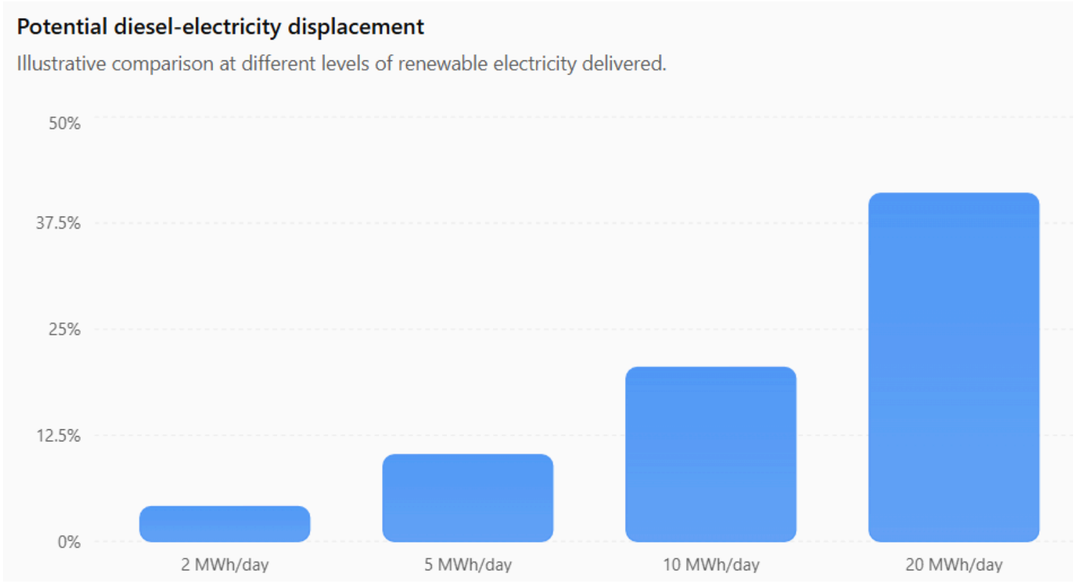
- **Warm seawater intake**: Pump warm surface seawater into a low-pressure flash-evaporation chamber.
- **Open-cycle OTEC power generation:** Under vacuum, part of the warm seawater flashes into low-pressure steam, which drives a low-pressure turbine connected to an electrical generator.
- **LTTD desalination integration**: Route suitable cold deep-seawater through a separate heat exchanger to condense water vapour and support low-temperature desalination.
- **Integrated output:** Collect the condensate as freshwater and use generated electricity to support seawater pumps, vacuum equipment and other auxiliary loads.
- **Installation cost:** Rs.15crores for 1 Lakh litres of freshwater.

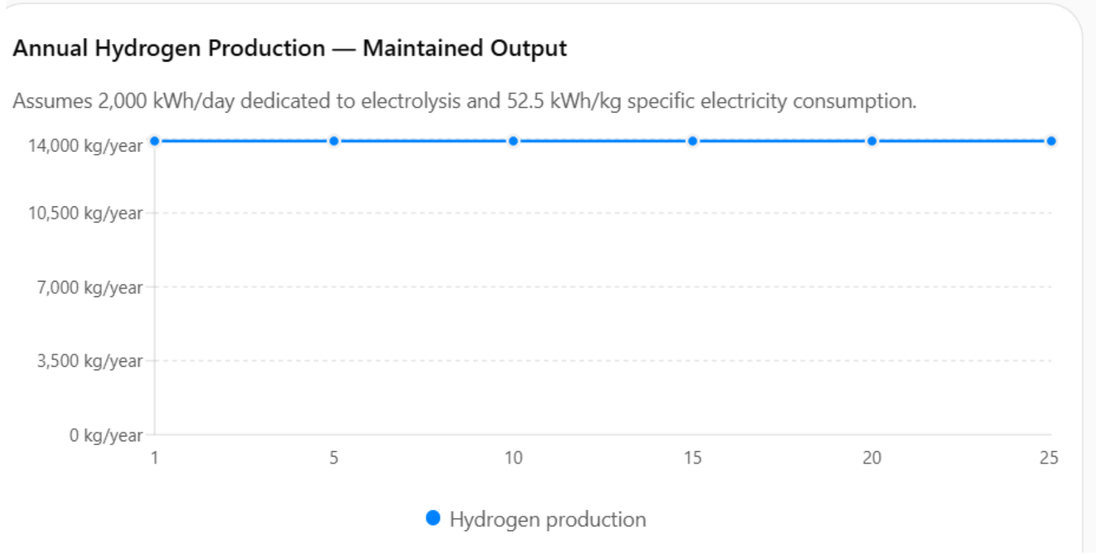


This system is self sustained in terms of power consumption as the power generated by it (around 528kWh/day) is fed to back the system and produce around 3.25 lakhs of litres of fresh water.

Green Hydrogen Production

1. **Renewable Integration:** Combine solar, wind, wave and OTEC energy through smart energy management.
2. **Hybrid Energy Storage:** Use batteries for short-term storage and green hydrogen for long-duration backup.
3. **Green Hydrogen Production:** Convert surplus renewable electricity into hydrogen using electrolysis.
4. **Energy Recovery:** Use stored hydrogen to generate electricity and reduce diesel dependence.
5. **Sustainability & Optimization:** Minimize conversion losses, costs and CO_2 emissions while ensuring safe storage.





IMPACT & BENEFITS

Environmental Impact

- Reduces diesel dependence and carbon emissions.
- Produces freshwater using ocean thermal energy.
- Integrates solar power for cleaner energy.
- Production of Green hydrogen.

SDG Goals





Economic Benefits

- Potentially lowers fuel and electricity costs.
- Combines energy generation with freshwater production.
- Enables scalable deployment across islands and coasts.

Social Impact

- Improves freshwater availability and reliability.
- Strengthens local energy and water security.
- Reduces dependence on Diesel fuel and watertransportation.
- Reduced Carbon Footprint by 4.1%.

Technical Benefits

- Integrates open-cycle OTEC with LTTD desalination.
- Utilises deep seawater for vapour condensation.
- Combines solar and ocean thermal energy.

RESEARCH & REFERENCES

SOLAR ENERGY:

1. Electricity Demand and Renewable-Energy Planning in Lakshadweep
2. Tariff reference for the proposed LTTD plant

WIND ENERGY::

1. Investigation of renewable energy potential in union territory of Lakshadweep islands
2. Official Electricity Tariff of Lakshadweep

WAVE ENERGY:

1. Characterization and verification of a two-body wave energy converter with a novel power take-off
2. Modelling and simulation of extreme wave heights around Agatti island of Lakshadweep, west coast of India

LTTD & OTEC:

1. Water quality improvement studies in LTTD plant
2. Experimental and feasibility study on the integration of OTEC with an LTTD desalination plant

SEASONAL CHANGES:

1. Seasonal solar-energy variation in Lakshadweep
2. Seasonal variation across renewable-energy sources
3. U.S.Department of Energy